Room Acoustic Measurement & Analysis Report

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1. Specifications

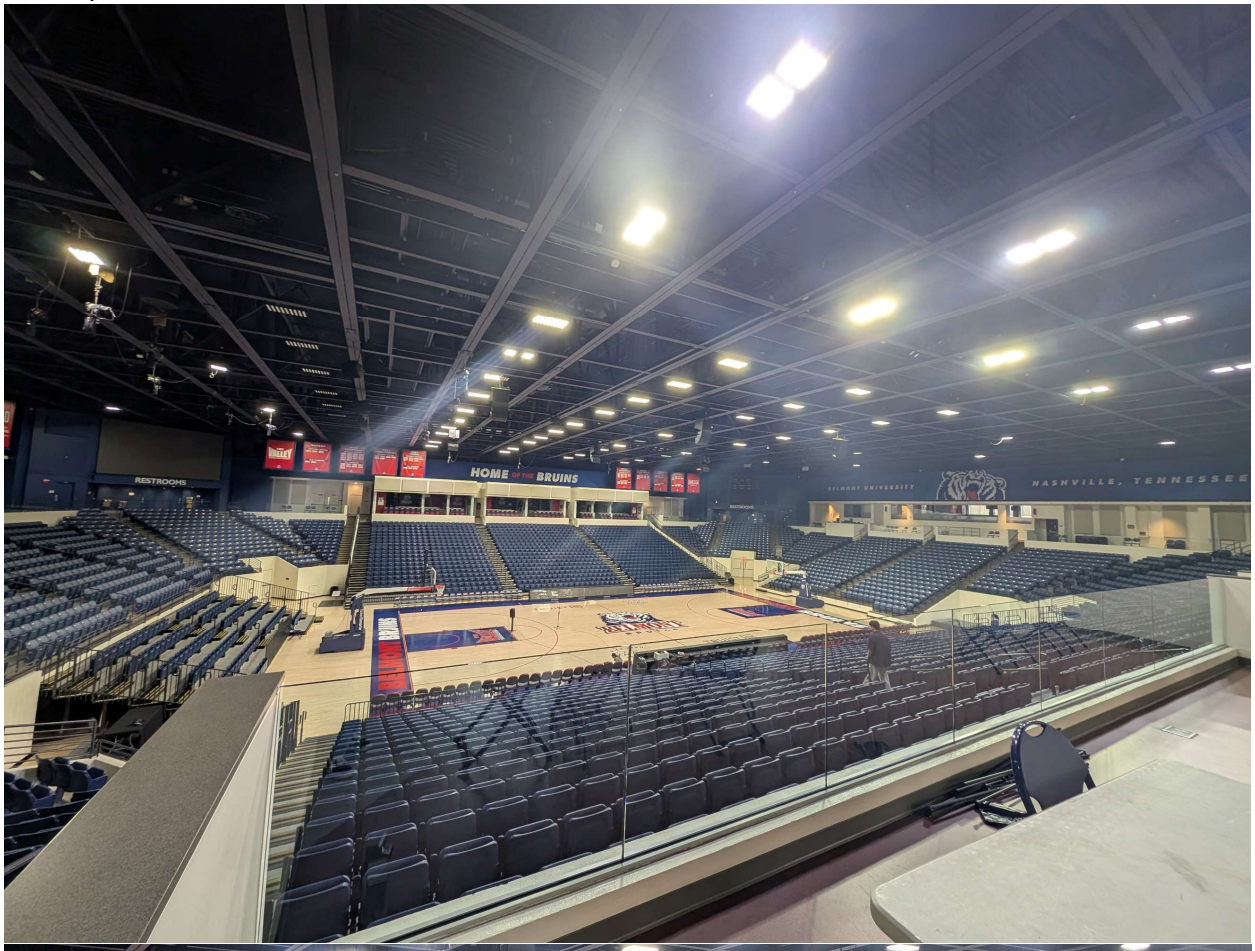
1.1. Curb Event Center Arena – used for athletics, concerts, meetings, conferences, etc.

1.2. Measurement date and time: 10am-12pm Friday, March 28, 2025

1.3. Temperature and humidity: 70deg F, 35% humidity

1.4. Background noise level (C-weighting, slow): 58db

1.5. Room photos:

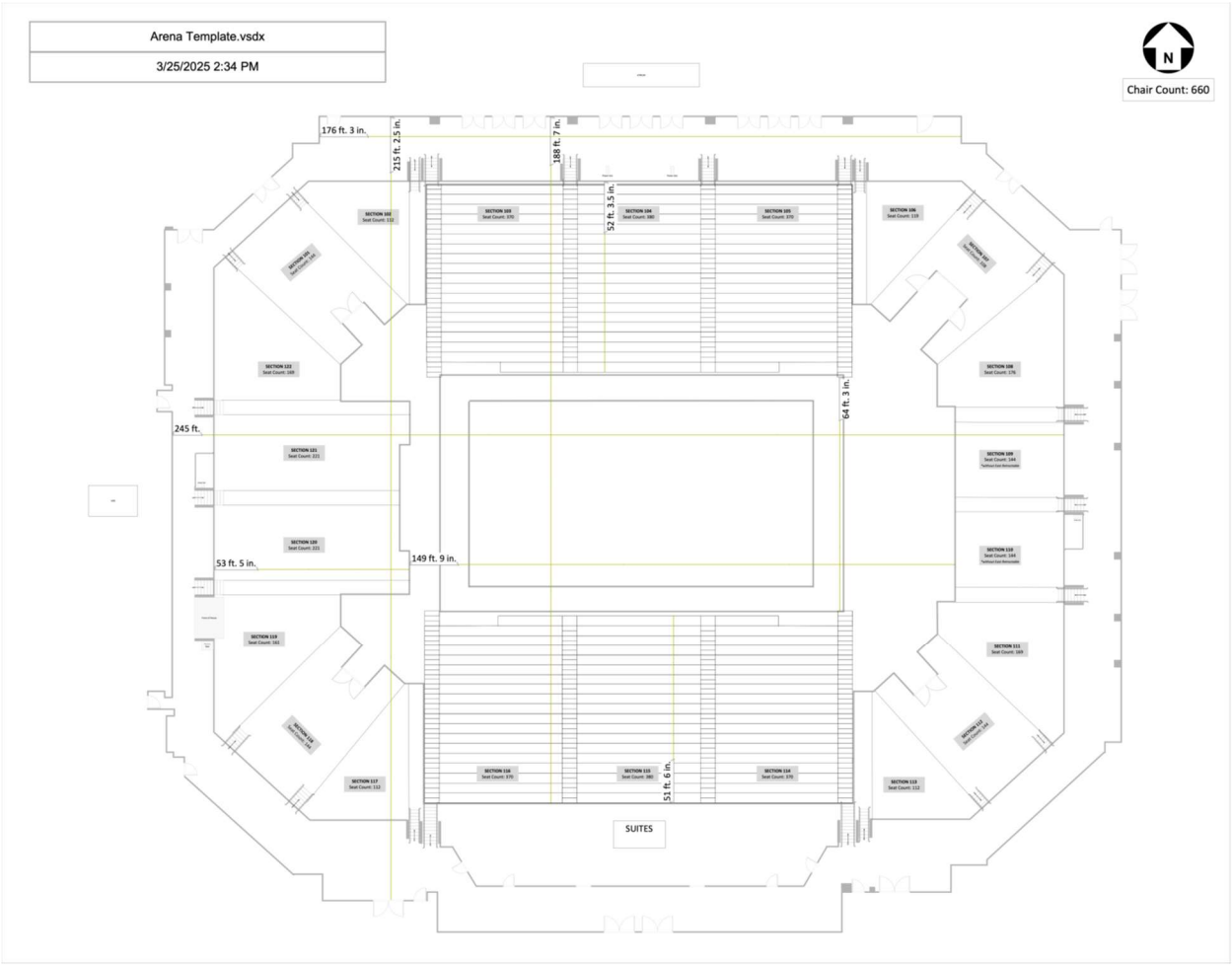




2. Reverb time estimation

2.1. Room dimensions: L = 245 ft, W = 215 ft, H = 58 ft

Floor plan sketch:



2.2. Estimated room materials:

Walls:
- Painted concrete, α = 0.06
- Sheetrock, α = 0.05
- Absorptive panels, Fiberglass board (2” thick), α = 0.99

Bleachers:
- Empty plastic/metal seats, α = 0.22

Stairs:
- Concrete, α = 0.015

Floor:
- Court (parquet on concrete), α = 0.07
- Bleachers, concourse (concrete), α = 0.015

Door, Box windows:
- Glass, α = 0.18

Ceiling:
-Steel, Metal deck, α = 0.99

Some absorption coefficients used from: https://www.acoustic-supplies.com/absorption-coefficient-chart/

2.3. Estimated RT60 at 500Hz using Sabine's formula:

Floor: 30% retractable bleachers, 20% court, 50% concrete seats
= (0.3 x 215 x 245 x 0.22) + (0.2 x 215 x 245 x0.07) + (0.5 x 215 x 245 x 0.015)
= 3476.55 + 737.45 + 395.0625
= 4609.0625

Wall 1 & 2 (the two short side walls):
20% absorptive panels, 60% concrete, 20% sheetrock
= (0.2 x 215 x 58 x 0.99) + (0.6 x 215 x 58 x 0.015) + (0.2 x 215 x 58 x 0.05)
= 2469.06 + 112.23 + 124.7
= 2705.99

Wall 3 (box suite side):
10% absorptive panels, 60% concrete, 30% sheetrock
= (0.1 x 245 x 58 x 0.99) + (0.6 x 245 x 58 x 0.015) + (0.3 x 245 x 58 x 0.05)
= 1406.79 + 127.89 + 213.15
= 1747.83

Wall 4:
10% absorptive panels, 40% door, 40% concrete, 10 % sheetrock
= (0.1 x 245 x 58 x 0.99) + (0.4 x 245 x 58 x 0.18) + (0.4 x 245 x 58 x 0.015) + (0.1 x 245 x 58 x 0.05)
= 1406.79 + 1023.12+ 85.26 + 71.05
= 2586.22

Ceiling: 100% metal deck
= 215 x 245 x 0.99
= 52148.25

Stairs: number of set of stairs = 8 metal, 18 concrete
rise of each step = 0.75 ft
raise of each step = 2 ft
number of steps per stairs = 40

= (8 x 0.75 x 2 x 40 x 0.99) + (18 x 0.75 x 2 x 40 x 0.015)
= 475.2 + 16.2
= 491.4

Total Absorption, A = 4609.0625 + 2705.99 + 2705.99 + 1747.83 + 2586.22 + 52148.25 + 491.4
= 66994.7425 sabine

Volume = L x W x H
= 245 x 215 x 58
= 3055150 cubic feet

RT60 = 0.049 x V / A
= 0.049 x 3055150 / 66994.7425
= 2.23 seconds

3. Measurement procedure

3.1. Loudspeaker, microphone, and audio interface specifications:

Loudspeaker – Mackie SRM350

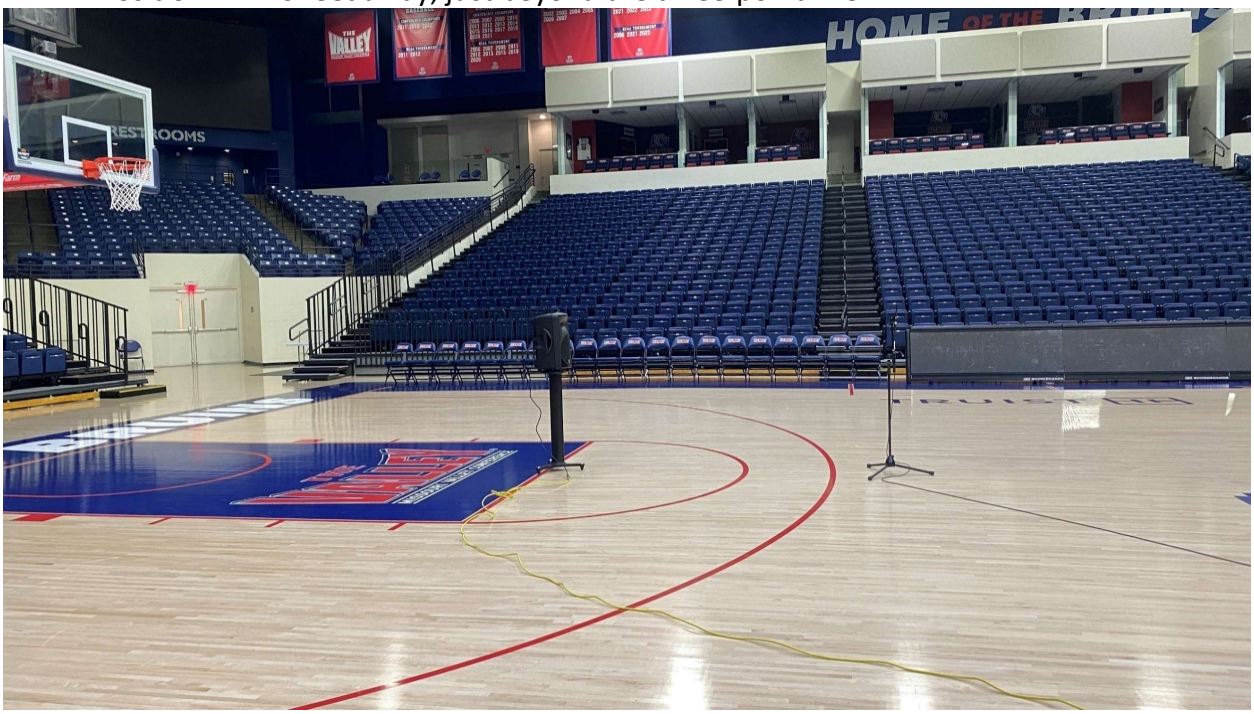
Frequency Response: (–3 dB) 83 Hz - 18.5 kHz
Sensitivity (center detent) Line: +4 dBu Mic: –36 dBu
THD: < 0.03%
Reference:
https://www.teletechnica.com/img/productos_multimedia/345a091432730e9189c572e8dfb1ca1c-archivo.pdf

Microphone - dbx RTA-M
Frequency Response: 20 Hz - 20 kHz
Sensitivity: -63 dB +-3 dB (0 dB=1V/ microbar 1,000 Hz indicated by open circuit)
Reference: https://dbxpro.com/en-US/products/rta-m

Audio Interface -
Frequency Response: 20 Hz to 20 kHz ±0.1 dB
Sensitivity:
THD+N: <0.002% (minimum gain, -1 dBFS input with 22 Hz/22 kHz bandpass filter)
Reference: https://fael-downloads-prod.focusrite.com/customer/prod/downloads/Scarlett%202i2%202nd%20Gen%20User%20Guide%20v1.1%20English%20-%20EN.pdf

3.2. SPL calibration: Calibrated to 85 dB SPL
3.3. REW measurement setup: Sweep length 512k, 2 repetitions, 44.1kHz, 6-second delay
3.4. Source and receiver positions

- Position 1— 10 feet away, just beyond the three-point line



- Position 2— 55.84ft away, at the other free throw line



- Position 3— 125 ft away; in the stands



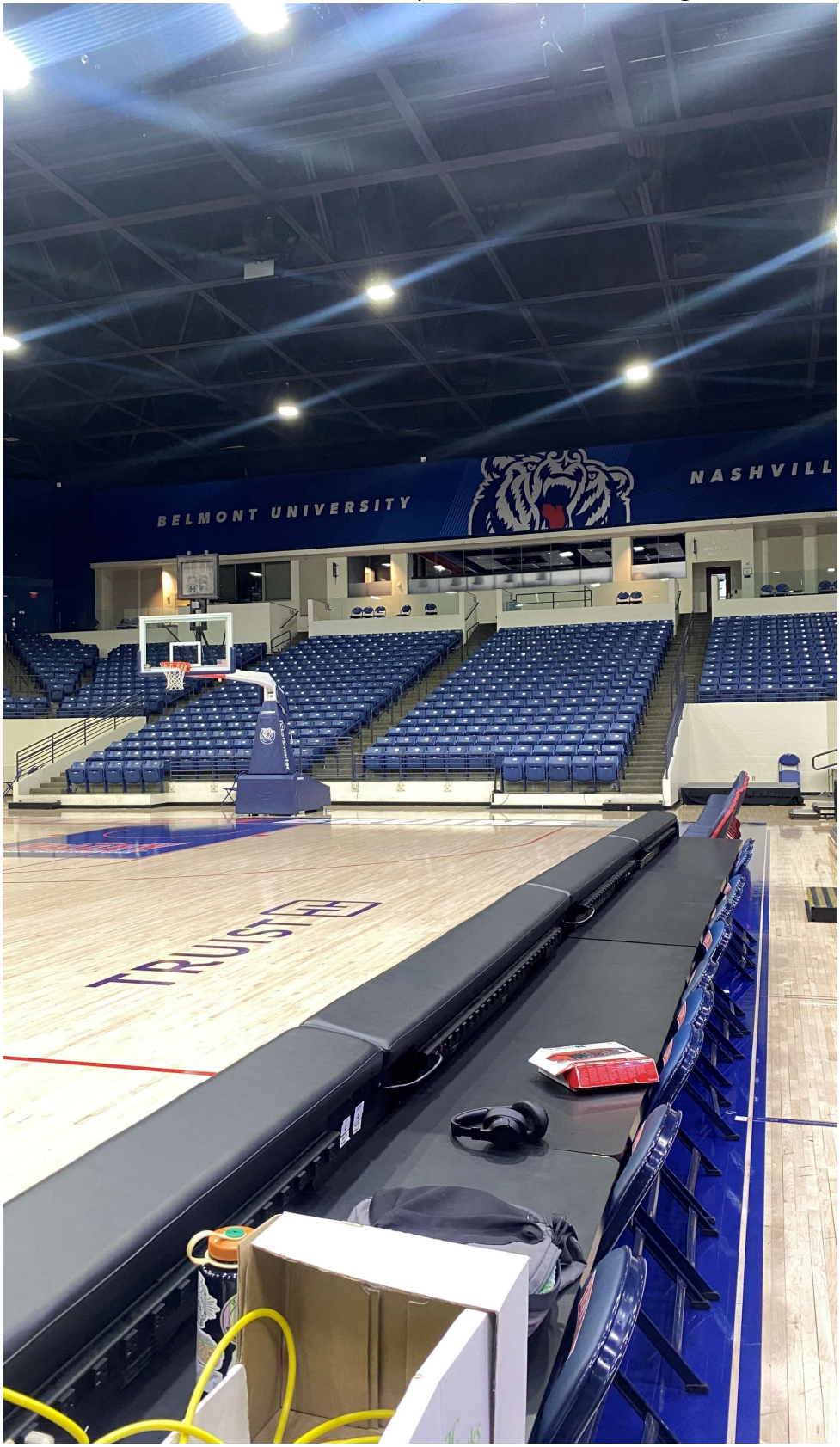
- Position 4— 41.83ft away; at the announcer/ref table on the far side of the court, even with the center line, to the left of the speaker



- Position 5— 93.62ft; suites level, about even with the free throw line of position 2 but to the left of the speaker

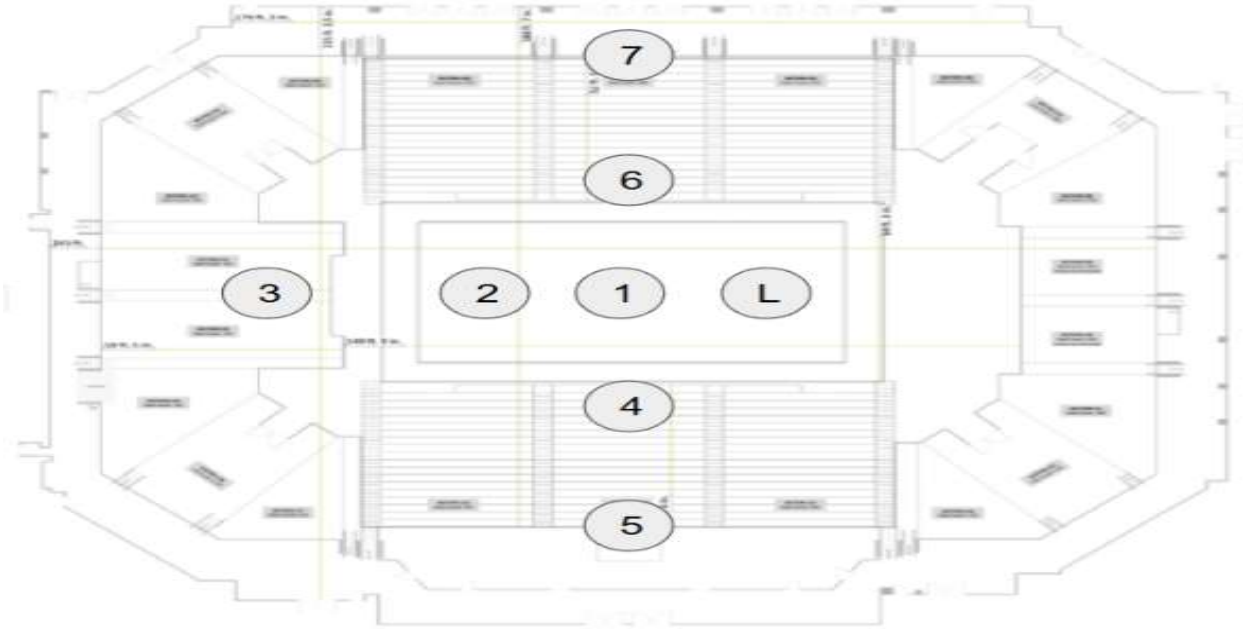


- Position 6— 44.25ft; same as position 4, but on the right side of the speaker



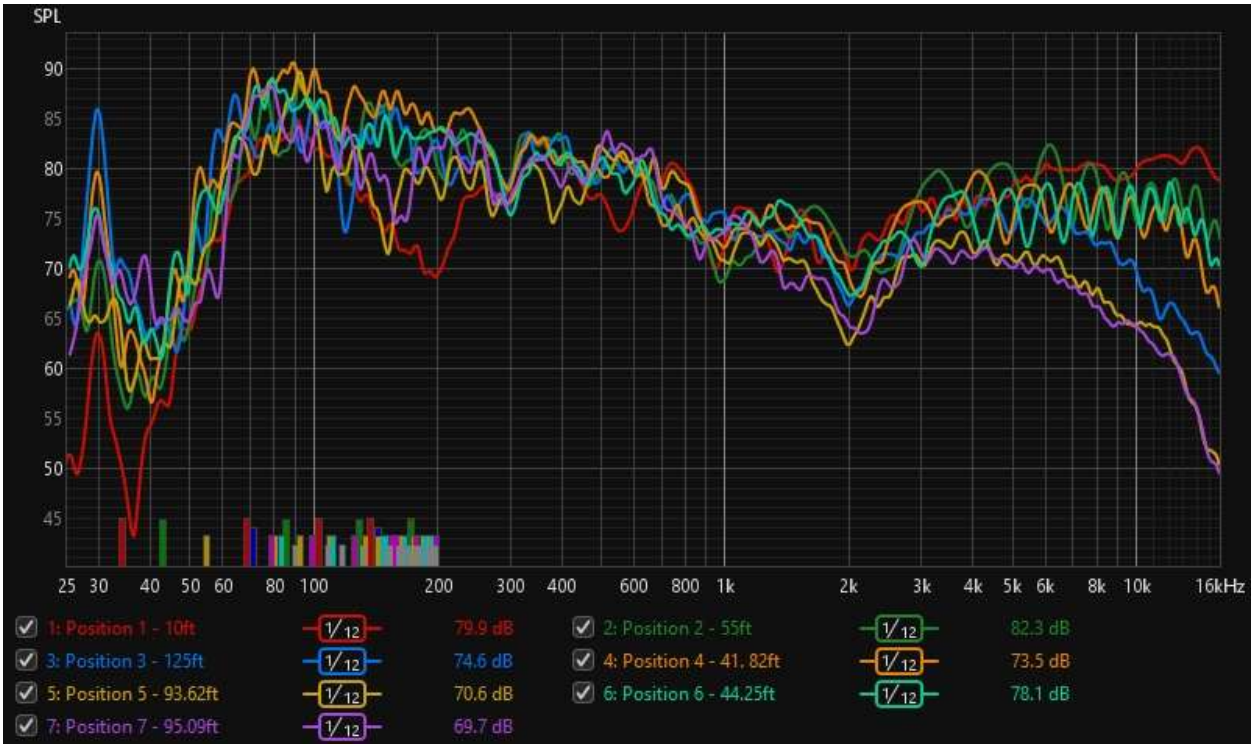
- Position 7—95.05ft; similar to position 5, at the concourse level on the right side of the speaker





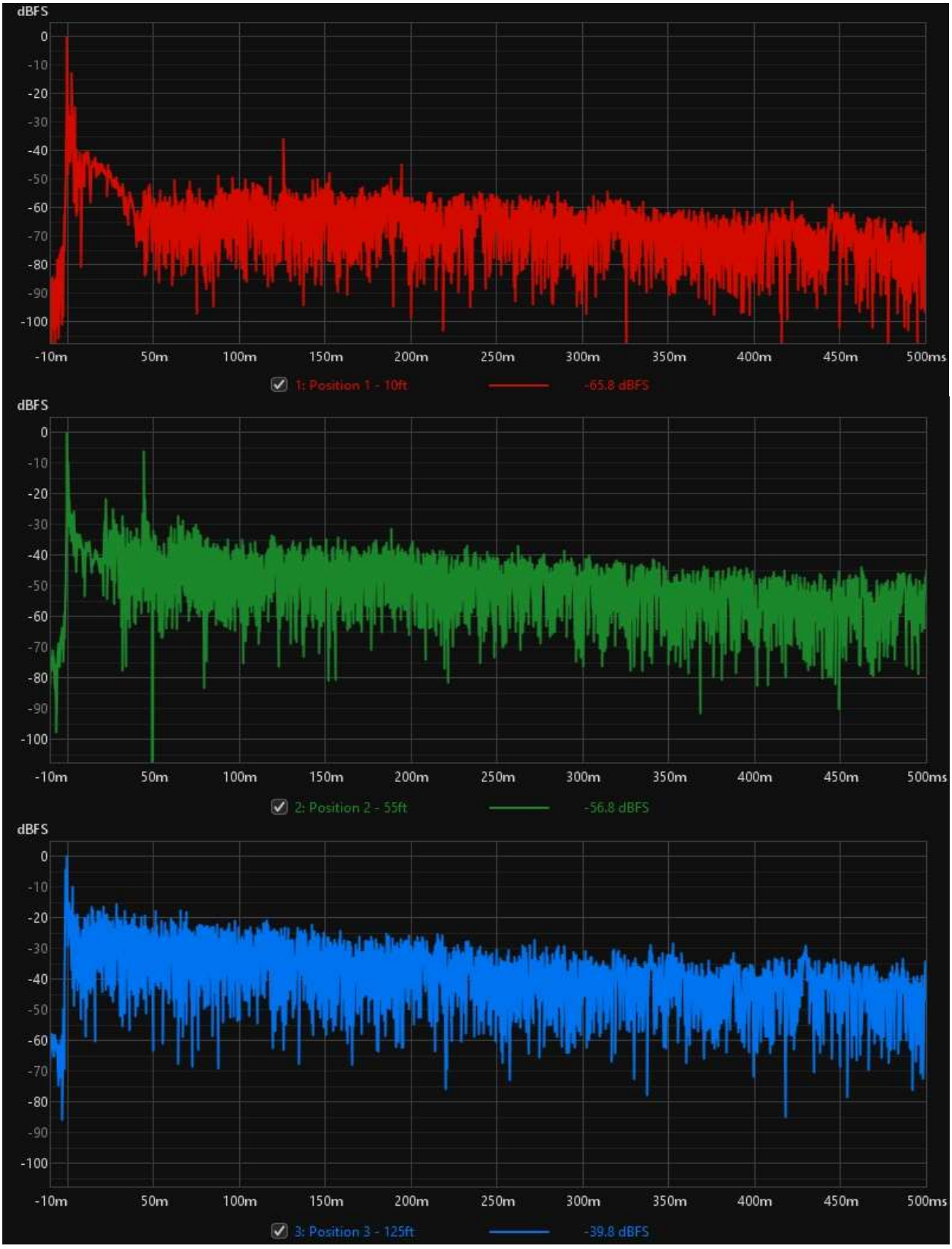
4. Measurement results

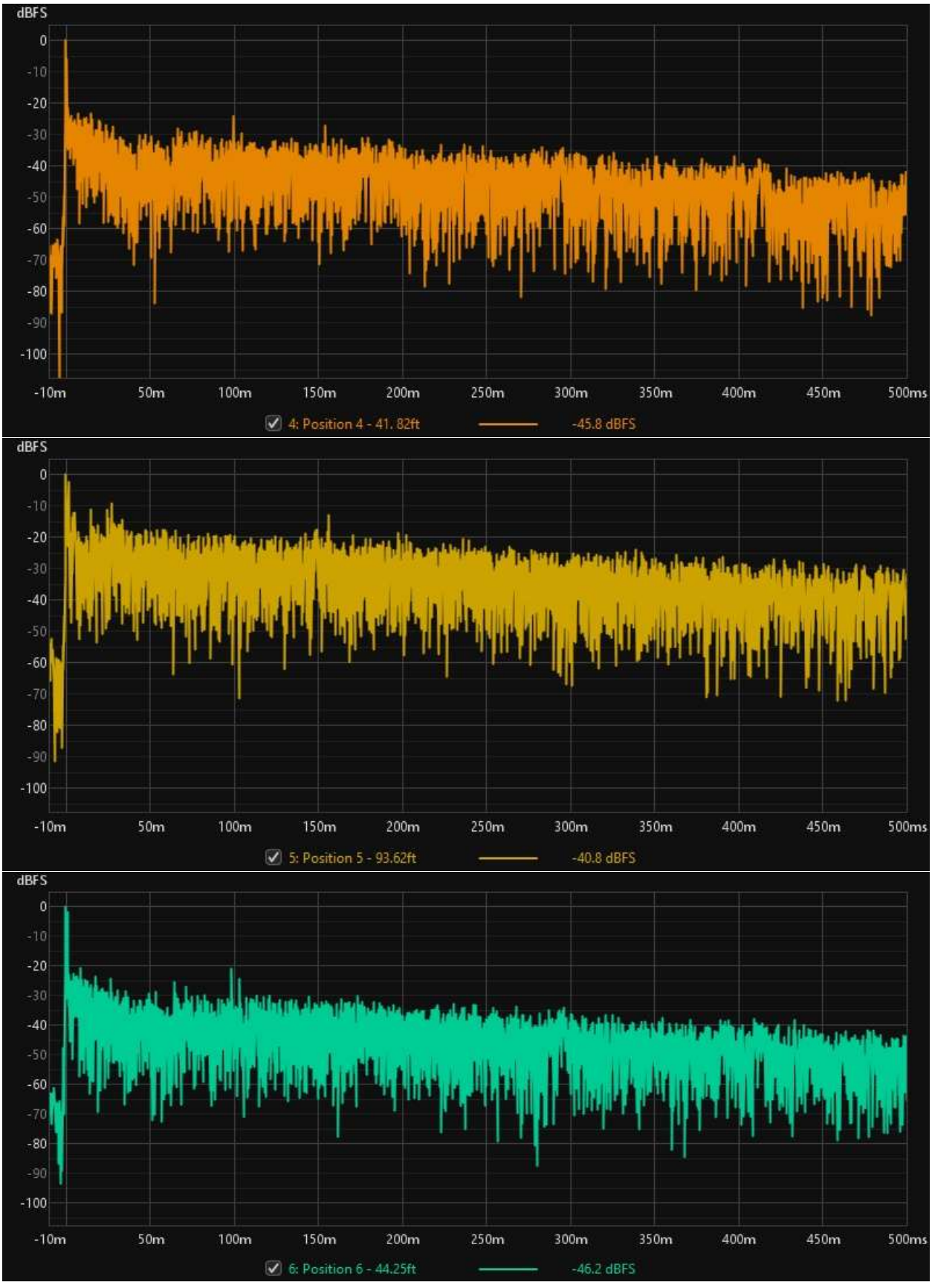
4.1. Magnitude (SPL) response

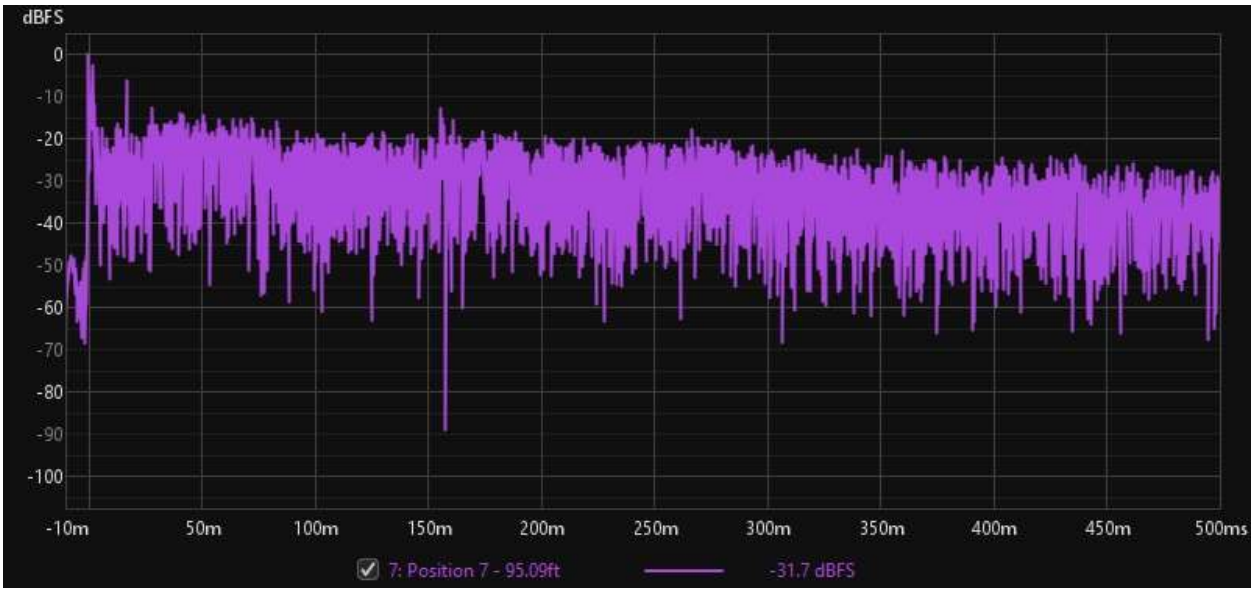


From the magnitude (SPL) response graph, we can see that low frequencies 25-60Hz have been subdued which can likely be attributed to the effect of absorptive panels all around the room. However, there is a big resonance at 30Hz for all positions except position 5 – 93.62ft. The first fundamental room mode is at 35Hz which justifies this big spike in resonance. Position 5 is located at the suites level, having better seats, so there might be better low frequency control for that area. For position 1, a 10 dB dip can be seen at around 200 Hz. From 200Hz onwards we can see similar trends for all seven positions. We see a significant dip of 5-10dB in frequency response at 2kHz. From comparing our results with other teams, we identified that this was likely due to the speaker's limitations rather than the acoustic properties of the room. Additionally, heavy comb filtering can be seen from 4 kHz onwards. This is most likely caused by the interaction between the direct sound from the speaker and the reflected sound from the shiny wooden basketball court.

4.2. Energy Time Curve (ETC)

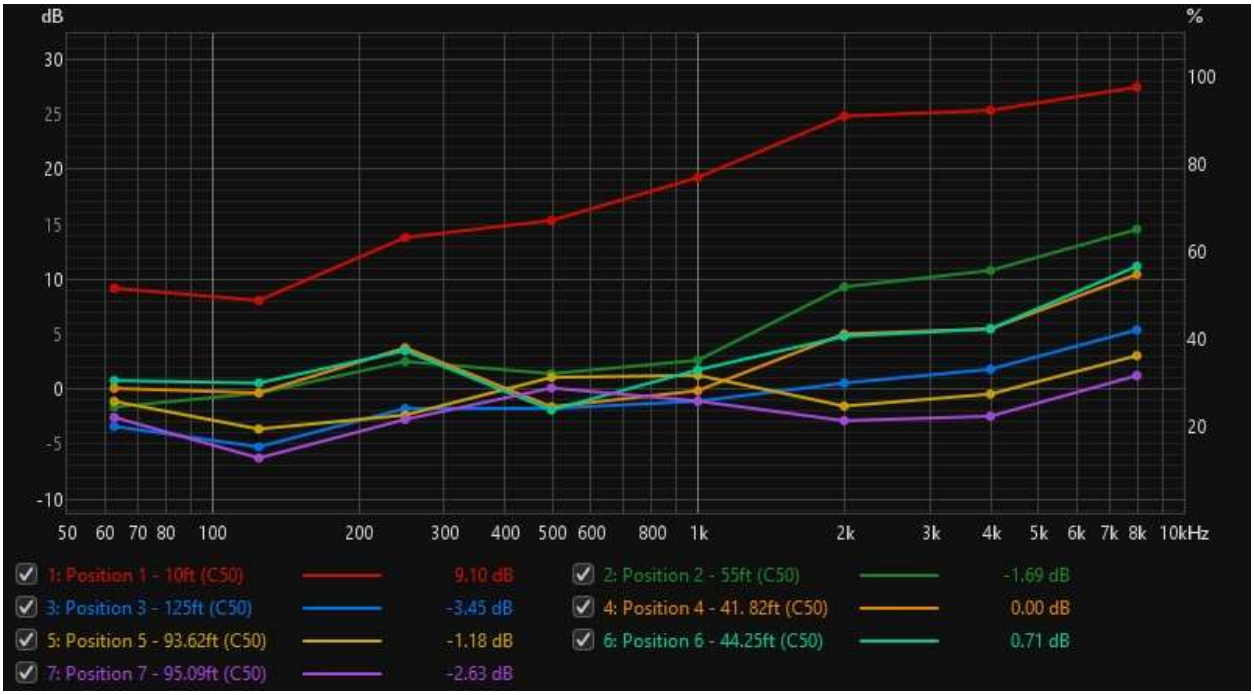



dBFS
0
-10
-20
-30
-40
-50
-60
-70
-80
-90
-100
-10m
50m
100m
150m
200m
250m
300m
350m
400m
450m
500ms
4: Position 4 - 41. 82ft
-45.8 dBFS
dBFS
0
-10
-20
-30
-40
-50
-60
-70
-80
-90
-100
-10m
50m
100m
150m
200m
250m
300m
350m
400m
450m
500ms
5: Position 5 - 93.62ft
-40.8 dBFS
dBFS
0
-10
-20
-30
-40
-50
-60
-70
-80
-90
-100
-10m
50m
100m
150m
200m
250m
300m
350m
400m
450m
500ms
6: Position 6 - 44.25ft
-46.2 dBFS



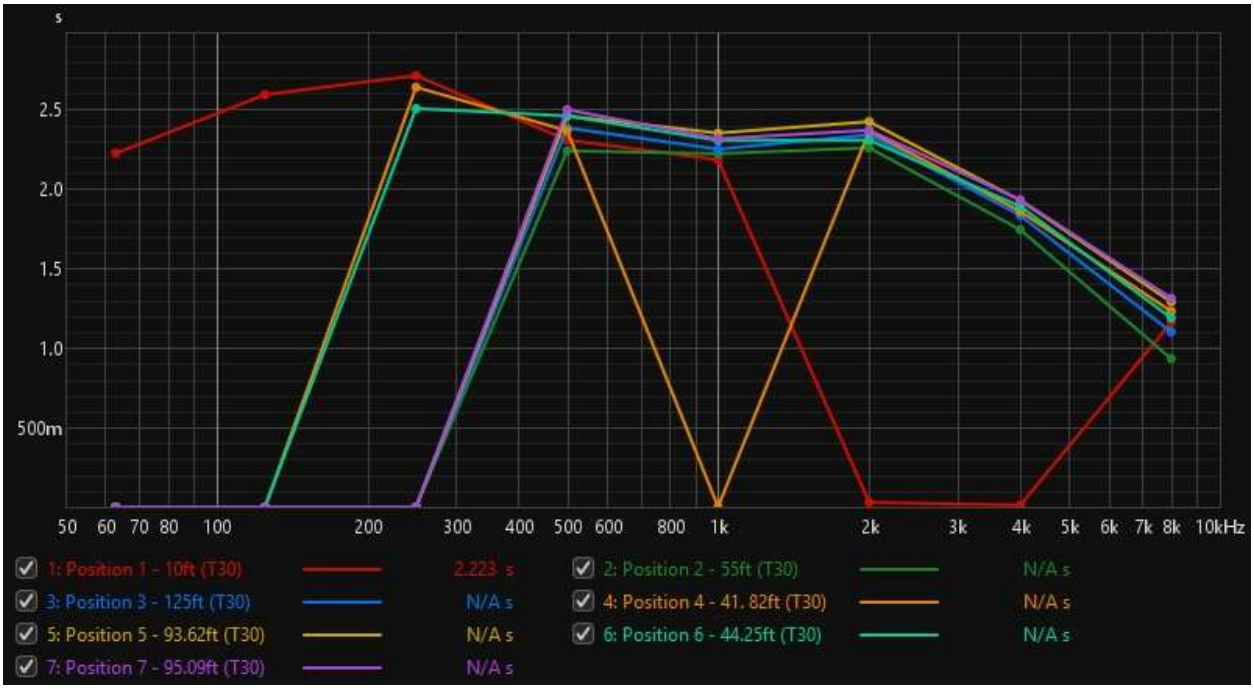
The Energy Time Curve (ETC) shows energy decay over time and can be used to detect early reflections. Since the noise floor was at 58 dB, we can see some energy even before 0ms for all positions. Moreover, there seems to be interaction between the noise floor and the sign sweep as there are several downward spikes. As for early reflections, we can see many upward spikes for position 1 directly after the initial transient after 0s which could be because of sound reflection off the court. There are also noticeable reflections at 125ms and 180ms. For Position 2, we can see a strong (almost the same energy as the first transient) early reflection at 50ms. For position 3, we can see that the first reflection was even stronger than the direct transient. Positions 4 displays a similar trend to position 1, with a strong reflection directly after the first transient and noticeable reflections at 100ms and 150ms. Position 5 is also like position 1, noticeable reflections are at around 25ms and 160ms. Similarly, position 6 has noticeable reflections at around 100ms. Same with position 7 with noticeable reflections at around 15ms and 160ms. For all positions, energy decay is slow and even after 500ms there is consistent energy which can be attributed to the level of the noise floor.

4.3. Clarity (C50)



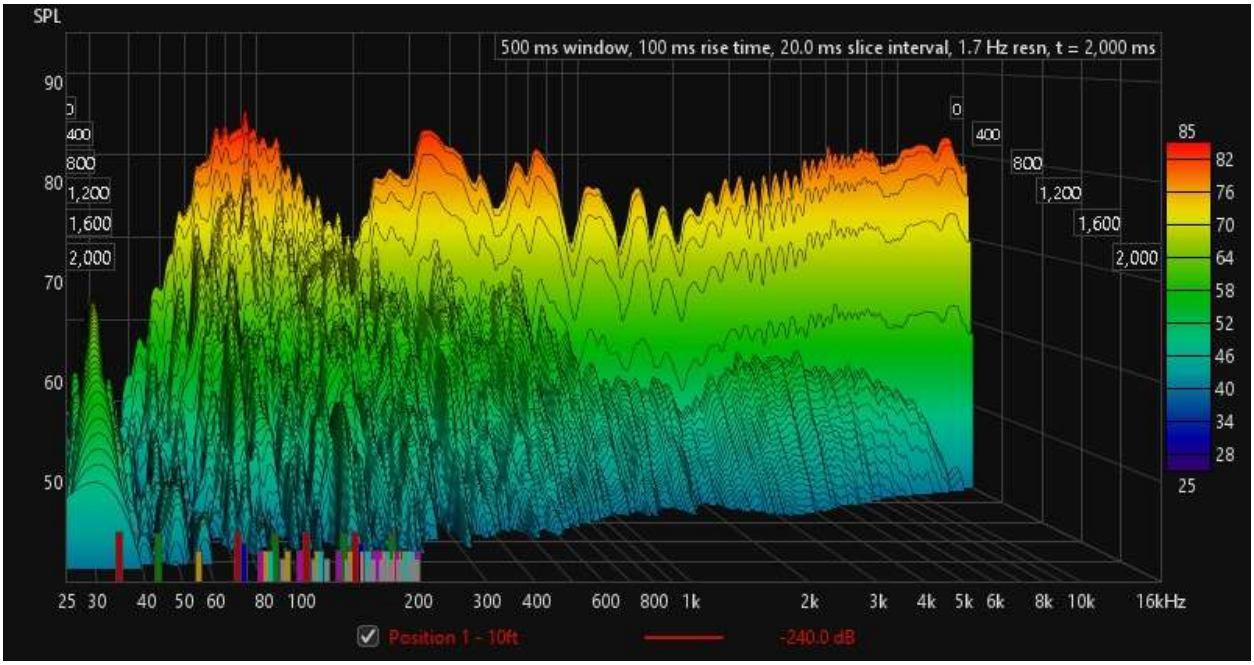
The clarity (C50) graph is the ratio of early sound energy within 50ms to late. A higher C50 value indicates better clarity. Position 1 had the highest clarity out of all positions because it was the closest to the loudspeaker. Moreover, higher frequencies 2kHz-8kHz have higher clarity than lower frequencies. This means that even at the position closest to the loudspeaker, early reflections muddy the sound which decreases the clarity value. This trend can also be seen for the other positions which imply that clarity of low and low-mid frequencies is not good. Human speech also lies in the low to mid-range so it can be very difficult to understand intelligibility of words if people are standing far from each other in the room. We experienced this firsthand when we were moving microphone positions, so the data supports this. From position 2-7, clarity gradually becomes worse as these positions are farther away from the loudspeaker.

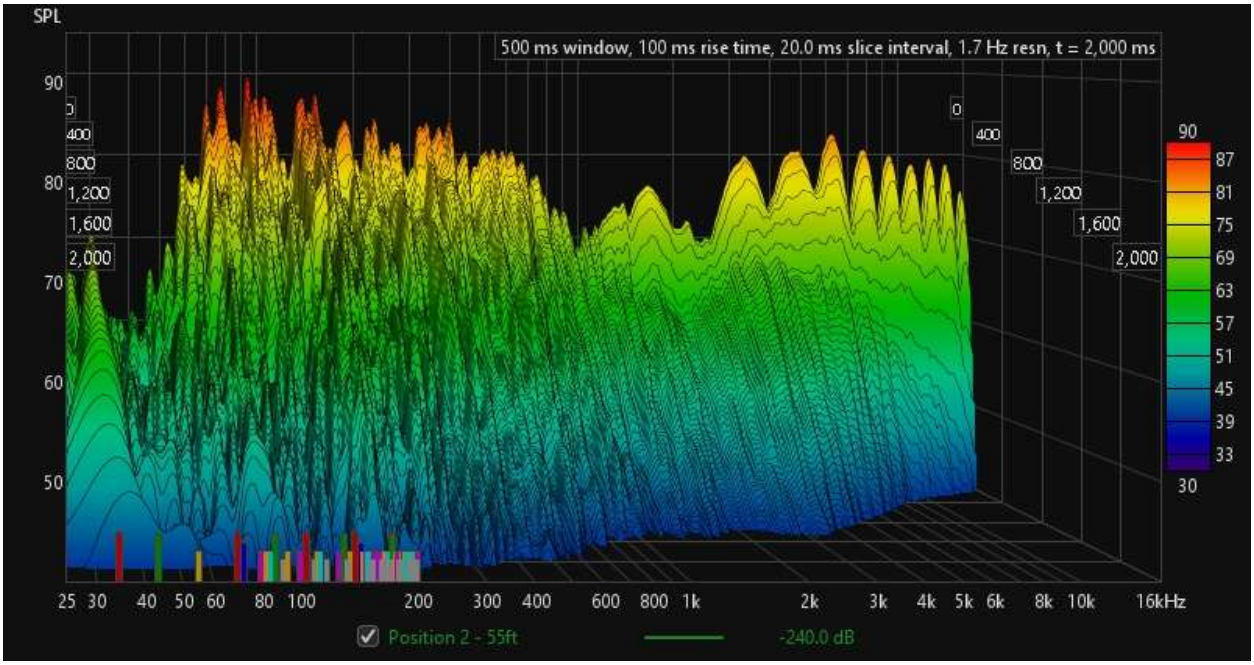
4.4. Reverb time (T30)

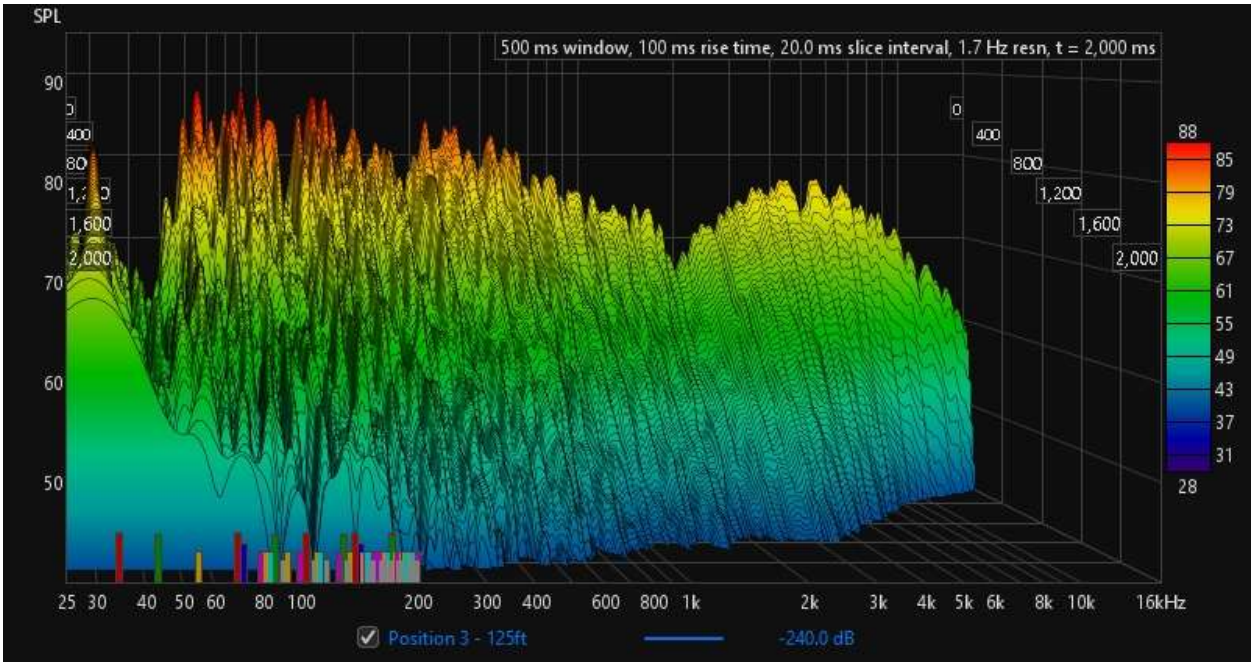


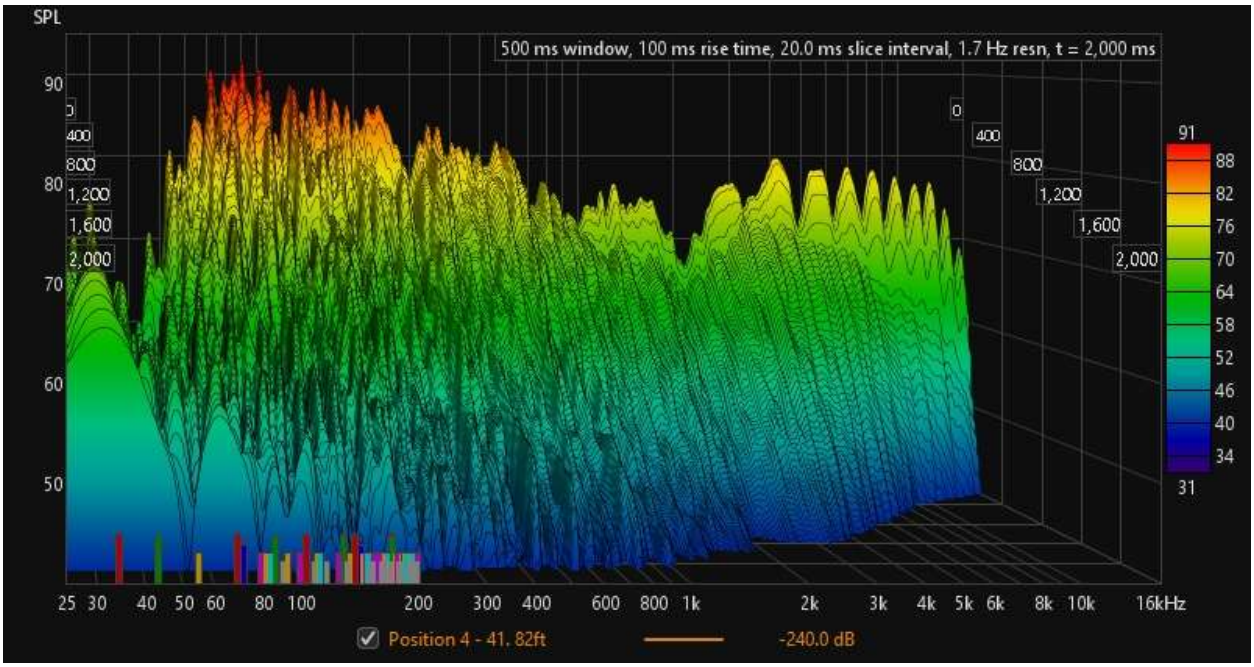
Reverberation time (T30) graph visually represents how long sound takes to decay by 30 dB. Higher T30 value represents longer reverberation and is suitable for big spaces like music halls. Whereas lower T30 value represents shorter reverberation and is better for speech clarity. From the above graph, we can see that T30 values differ based on frequency. Position 1 showed the highest T30 values from 2.23s-2.70s for frequencies from 60Hz-250Hz. And T30 values all gradually decrease for all positions at higher frequencies. For positions 2 through 7, the T30 is fairly consistent from 2.23s-2.50s from 200Hz-2kHz. Anomalies can be seen at places where the software failed to calculate the reverb time resulting in a value of 0 seconds for reverb time. This means that the room has a good reverb time when used for athletics and performances, but it may not be ideal when hosting conferences and meetings.

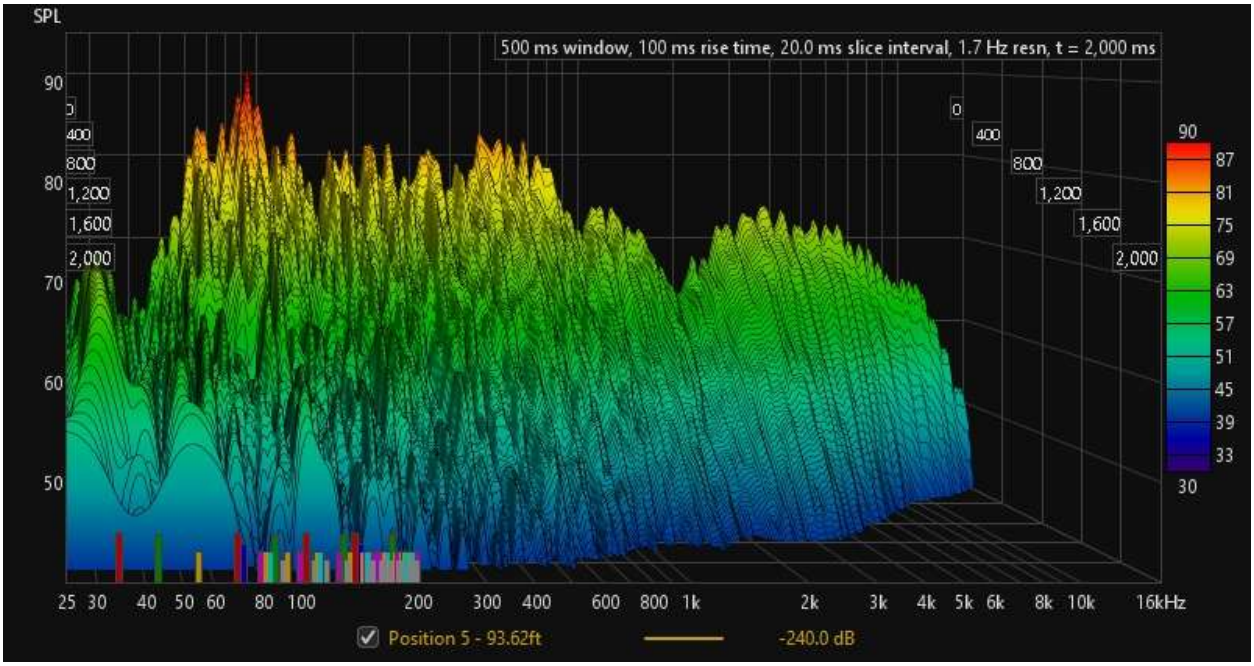
4.5. Waterfall diagrams

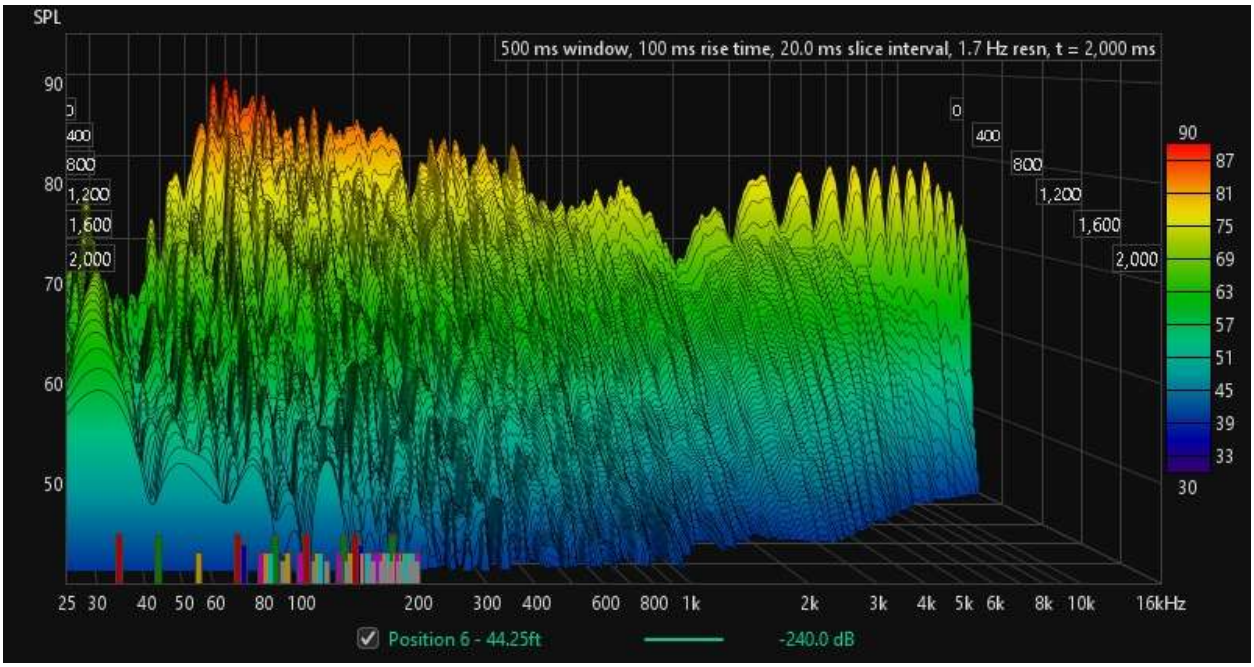


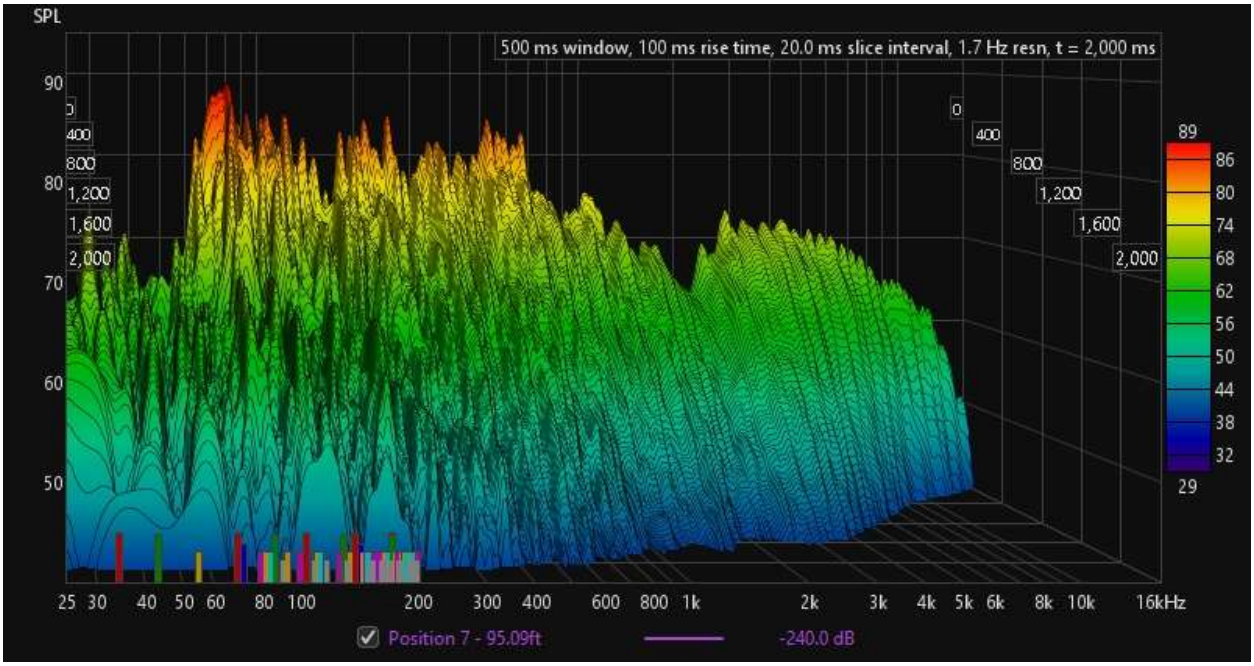



SPL
500 ms window, 100 ms rise time, 20.0 ms slice interval, 1.7 Hz resn, t = 2,000 ms
90
80
70
60
50
0
400
800
1,200
1,600
2,000
88
85
79
73
67
61
55
49
43
37
31
28
25 30 40 50 60 80 100 200 300 400 600 800 1k 2k 3k 4k 5k 6k 8k 10k 16kHz
Position 3 - 125ft
-240.0 dB


SPL
500 ms window, 100 ms rise time, 20.0 ms slice interval, 1.7 Hz resn, t = 2,000 ms
90
80
70
60
50
0
400
800
1,200
1,600
2,000
91
88
82
76
70
64
58
52
46
40
34
31
25 30 40 50 60 80 100 200 300 400 600 800 1k 2k 3k 4k 5k 6k 8k 10k 16kHz
Position 4 - 41. 82ft
-240.0 dB


SPL
500 ms window, 100 ms rise time, 20.0 ms slice interval, 1.7 Hz resn, t = 2,000 ms
90
80
70
60
50
0
400
800
1,200
1,600
2,000
90
87
81
75
69
63
57
51
45
39
33
30
25 30 40 50 60 80 100 200 300 400 600 800 1k 2k 3k 4k 5k 6k 8k 10k 16kHz
Position 5 - 93.62ft
-240.0 dB


SPL
500 ms window, 100 ms rise time, 20.0 ms slice interval, 1.7 Hz resn, t = 2,000 ms
90
80
70
60
50
0
400
800
1,200
1,600
2,000
90
87
81
75
69
63
57
51
45
39
33
30
25 30 40 50 60 80 100 200 300 400 600 800 1k 2k 3k 4k 5k 6k 8k 10k 16kHz
Position 6 - 44.25ft
-240.0 dB



The waterfall diagrams help visualize the decay of energy of different frequencies across time. For position 1, starting at 300ms, the decay for frequencies after 500 Hz was very rapid. Most of the mid to high frequencies die around 800ms which shows that reverb time for these frequencies at this position is very low. For position 2-7 mid to high frequencies die at around 1200ms which is in line with the reverb graph showing consistent reverb time for these positions. Multiple spikes can be seen at low frequencies which indicate several echoes for all positions which is an adverse effect of reflections. Moreover, these spikes are more prominent at room modes like 35Hz, 43Hz, 70Hz, and so on. The SPL levels of several echoes are high even after 300ms which indicate that the room will sound very muddy. After 400ms, frequencies between 2.5 kHz and 6 kHz seem to take a longer time to decay than frequencies at 2 kHz and 8 kHz which indicate some sustained ringing. The dip at 2k is also likely because of the limitation of the loudspeaker. Positions 3 through 7 show consistent patterns, with steady decay above 500 Hz and persistent low-frequency echoes at various times.